



Significant Figures: Precision & Measurement Mastery

Rules, Avoidance of Ambiguity, and Operations

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Session Flow: Significant Figures (100 min)

A 100-minute student-centered sequence based on Buckingham's PGC Framework.

Lesson Structure Overview

- **15 min – Phase 1: Research & Classroom Submission** (Independent Inquiry & Upload)
- **30 min – Phase 2: Teacher Clarification & Deep Dive** (Rules, Trailing Zeros & Video Analysis)
- **35 min – Phase 3: Effort & Scaffolding** (4-Level Workshop & Error Analysis)
- **20 min – Phase 4: Recognition & Synthesis** (Peer Assessment Protocol & Reflection)

1. Novelty & Curiosity | Research Task & Google Classroom Submission

Independent Research Challenge (15 min) – GRADED TASK

Search online using your laptop/tablet and create a brief, nicely formatted note document addressing the following four questions:

1. **What** are Significant Figures (Sig Figs)?
2. **Why** do we use them? (What is their purpose in science/math?)
3. **How** do they help us avoid confusion in real-world measurements?
4. **What** are the fundamental rules to count them?
(Include at least one example for each rule)
5. **How** does Scientific Notation work, and how does it eliminate zero ambiguity in Sig Figs?
6. **What** happens when we calculate with Sig Figs? Explain the rules for:
 - Multiplication and Division (limiting by least number of sig figs)
 - Addition and Subtraction (limiting by least number of decimal places)

Submission Requirements

- Format your document clearly. Use **Canva** or **Google Slides**.
- Export as PDF and upload directly to **Google Classroom**.
- **Note:** This activity counts toward your classwork grade!

2. Guided Discovery | What, Why, and How?

What are Significant Figures?

The digits in a measured quantity that carry meaningful precision based on the limits of the measuring device.

Why do we use them? A Life-or-Death Example

Imagine a laboratory formulating a high-precision medication, where a dose above 0.050 g (50 mg) of an active compound is lethal. If a technician uses a standard kitchen scale (measures to 0.1 g):

- Placing 0.049 g (**safe**) displays: **0.0** g (rounds down). Placing 0.051 g (**lethal**) displays: **0.1** g (rounds up).
- **Reality check:** A tiny difference of 0.002 g changes what the screen shows, but the scale hides the actual digits!

Using an analytical balance precise to four decimal places:

- The scale displays: **0.0482** g (3 significant figures).
- Consequence: We know with physical certainty that the dose is safely below the lethal threshold.

Key Takeaway: Significant figures reflect how many digits in a measurement can actually be trusted in the real world.

2. Guided Discovery | What, Why, and How?

How do they help us avoid confusion in real-world measurements?

Imagine a chemist writes down a safe dose of **2.5** g on a lab report, but another technician needs to prepare it.

- Did they weigh it as **2.5** g using a rough scale? (Actual mass could be 2.46 g or 2.54 g).
- Or was it measured to exactly **2.500** g on a high-precision balance?

Without knowing the instrument's limits, the team cannot tell how trustworthy that single trailing zero is. Writing **2.500** g explicitly communicates high precision, ensuring no one guesses whether the value was rounded.

Video Resource

Watch this video to review the rules for counting significant figures:

[Click here to open Video](#)

2. Guided Discovery | The General Rules

Rule 1: Non-Zero Digits

All non-zero digits are **always significant**.

$148.3 \Rightarrow 4 \text{ sig figs}$

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Rule 2: Captive (Sandwiched) Zeros

Zeros between non-zero digits are **always significant**.

$$1004 \implies 4 \text{ sig figs} \quad | \quad 5.008 \implies 4 \text{ sig figs}$$

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Rule 3: Leading Zeros

Zeros to the left of the first non-zero digit are **NEVER significant** (they are placeholders).

$$0.0025 \implies 2 \text{ sig figs (2 and 5)}$$

2. Guided Discovery | Deep Dive: Trailing Zeros

Rule 4: Trailing Zeros | Why are Trailing Zeros so confusing?

Zeros at the end of a number have a dual role:

1. Are they just **placeholders** to show how big the number is?
2. Or were they **actually measured** by a precision tool?

2. Guided Discovery | Deep Dive: Trailing Zeros

Rule 4: Trailing Zeros | Why are Trailing Zeros so confusing?

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Case A: Decimal Point PRESENT

Trailing zeros **ARE** significant.

- Writing 12.00 g means the tool measured down to .01 and got .00.
- **Example:** 45.00 $\implies$ 4 sig figs
-
- **Example:** 400. $\implies$ 3 sig figs (*The decimal point explicitly marks all zeros as significant*).

Case B: NO Decimal Point

Trailing zeros are **NOT** significant (placeholders/ambiguous).

- Writing 2500 m usually means rounded to the nearest hundred.
- **Example:** 2500 $\implies$ 2 sig figs

2. Guided Discovery | Master Table: Avoiding Zero Ambiguity

Measurement	Decimal?	Sig Figs	Why / Explanation
5000	NO	1	Zeros only locate the magnitude (placeholders).
5000.	YES (at end)	4	Explicit dot forces all zeros to count.
5.000	YES	4	Zeros after decimal reflect fine precision.
0.0050	YES	2	Leading zeros fail; trailing zero passes.

The Ultimate Golden Solution

When in doubt, write numbers in **Scientific Notation**! It isolates precision from magnitude.

2. Guided Discovery | The Golden Rule: Scientific Notation

The Ultimate Golden Rule

Scientific notation ($N \times 10^k$) removes all zero ambiguity by separating **precision** (in N) from **magnitude** (in 10^k).

Example: How to write 5000 with exact precision

Scientific Form	Sig Figs	What it tells us about precision
5×10^3	1	Measured to the nearest thousand (5000 ± 500).
5.0×10^3	2	Measured to the nearest hundred (5000 ± 50).
5.00×10^3	3	Measured to the nearest ten (5000 ± 5).
5.000×10^3	4	Measured to the nearest unit (5000 ± 0.5).

Takeaway Rule

Count **all** digits in the coefficient (N). Ignore the power of ten (10^k) completely!

2. Guided Discovery | Target 3: Calculations & Rounding

Multiplication & Division

Round final answer to the **fewest total sig figs** of any number in the calculation.

$$4.56 \times 1.4 = 6.384$$



6.4 (Restricted by 1.4 $\Rightarrow$ 2 sig figs)

Addition & Subtraction

Round final answer to the **fewest decimal places** (least precise position).

$$12.11 + 0.3 = 12.41$$



12.4 (Restricted by 0.3 $\Rightarrow$ 1 d. p.)

3. Effort & Scaffolding | Level 1: Identification (8 min)

[Level 1: Basic – Individual Work]

State the number of significant figures in each measurement.

1. 0.00405 kg $\Rightarrow$

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[Level 1: Basic – Individual Work]

State the number of significant figures in each measurement.

1. 0.00405 kg $\Rightarrow$ 3 sig figs

2. 120.00 mL $\Rightarrow$

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[Level 1: Basic – Individual Work]

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1. 5000 m $\Rightarrow$ 1 sig fig (Ambiguous)

2. 3.080×10^4 s $\Rightarrow$

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1. 5000 m $\Rightarrow$ 1 sig fig (Ambiguous)

2. 3.080×10^4 s $\Rightarrow$ 4 sig figs

3. Effort & Scaffolding | Level 2: Error Analysis (9 min)

[Level 2: Medium – Pair Work]

Spot the error in precision and correct it.

Spot the Error!

A student calculates:

$$12.55 \text{ m} \times 2.1 \text{ m} = 26.355 \text{ m}^2$$

and leaves 26.355 m^2 as their final answer.

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and leaves 26.355 m^2 as their final answer.

Correction

Multiplication rule applies: 2.1 has only 2 sig figs.

Correct Answer: 26 m^2 .

3. Effort & Scaffolding | Level 3 & 4: Master Challenges (18 min)

[Level 3: Hard]

Calculate $(15.43 - 12.1) \div 1.25$ respecting operations precedence and sig fig rules.

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Solution

1) $15.43 - 12.1 = 3.33 \rightarrow 3.3$ (1 decimal place = 2 sig figs)

2) $3.3 \div 1.25 = 2.64 \rightarrow \mathbf{2.6}$ (2 sig figs)

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Solution

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[Level 4: Expert Challenge]

Express 450000 with exactly 3 significant figures.

$$\mathbf{4.50 \times 10^5}$$

4. Recognition & Feedback | Peer Assessment Protocol

Notebook Exchange Protocol – Level 3 (7 min):

Layer 1 (Praise): Highlight clear application of rules and correct identification of zero status.

Layer 2 (Critique): Check for improper rounding or mixing multiplication vs. addition rules.

Layer 3 (Validation): Sign and return notebook to confirm peer accuracy.

Zero Type	Status	Example
Leading Zeros	Never Significant	0.003 → 1 sig fig
Captive Zeros	Always Significant	4002 → 4 sig figs
Trailing (with decimal)	Significant	45.00 → 4 sig figs
Trailing (no decimal)	Ambiguous / Placeholders	200 → 1 sig fig

Master precision in measurement!